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Efficacy and Pharmacokinetics of 3 Sustained-Release Buprenorphine Formulations in Cairo Spiny Mice (*Acomys cahirinus*)

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African spiny mice (*Acomys* spp.) are an emerging animal model for regeneration due to their remarkable healing capabilities. Defining characteristics of these species include fragile skin that sloughs easily and shedding of the tail skin when grabbed, making handling and administration of parenteral drugs difficult in conscious animals. In addition, many studies in spiny mice involve painful procedures. To our knowledge, there are no reports regarding analgesia in *Acomys* spp. This prospective study aimed to assess 3 sustained-release buprenorphine formulations—lipid-bound extended-release buprenorphine (XRB), polymeric sustained-release buprenorphine (SRB), and long-acting transdermal buprenorphine (TB)—in spiny mice. Adult male and female Cairo spiny mice (*Acomys cahirinus*) were included and administered 1 of the 3 treatments (XRB 3.25 mg/kg, SRB 1 mg/kg, or TB 20 mg/kg). Tail flick assays to assess nociception and terminal blood collection for pharmacokinetic analysis were performed at baseline and a set timepoint following treatment administration (1, 2, 4, 8, 24, 48, and 72 hours) (n = 3 of each sex per timepoint per treatment group). Additional animals underwent repeated serial tail flick assays at 1, 2, 4, 8, 24, 48, and 72 hours following the administration of treatment or sterile saline (n = 5-8 of each sex per treatment group). XRB displayed the longest duration at which mean plasma buprenorphine concentrations were >1 ng/mL, with the drug remaining above this level for 48-72 hours, compared with 24-48 hours in spiny mice receiving TB and 8 hours in those receiving SRB. On serial tail flick assays, the mean percentage of maximum possible efficacy was highest at all timepoints up to 72 hours in the TB group, followed by the XRB group, suggestive of greater analgesic efficacy. Several spiny mice receiving SRB developed ulcerative skin lesions within 24 hours of administration, so this treatment is not recommended in *Acomys* without additional evaluation. XRB and TB are promising analgesic therapies in spiny mice, and treatment selection should be based on whether duration of action or ease of application is the priority.

Abbreviations and Acronyms: $AUC_{0\text{-inf}}$, AUC extrapolated to infinity; Bup-SR, buprenorphine sustained release; %MPE, percentage of maximum possible efficacy; SRB, polymeric sustained-release buprenorphine; TB, long-acting transdermal buprenorphine; XRB, lipid-bound extended-release buprenorphine

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Introduction

The African spiny mouse (*Acomys* spp.) is an increasingly popular animal model used in biomedical research. Historically used to model type 2 diabetes mellitus, these animals have also been used to research menstruation, pre- and perinatal development, and, most recently, regeneration.[1] They are a promising model for regeneration due to their remarkable healing capabilities, which have been described in several tissues including the skin, muscles, heart, kidneys, and spinal cord. Many of these tissues exhibit scarless healing in response to injury, setting these animals apart from other rodent species.[2]

Despite their increasing use in a laboratory setting, spiny mice possess some characteristics that make handling difficult. They have particularly fragile skin that sloughs easily, with a previous study having found that the tensile strength of mouse (*Mus musculus*) skin was 20-fold greater than that of *Acomys percivali* or *Acomys kempi*.[3] In addition to this, spiny mice also exhibit false caudal autotomy in which the tail skin sheath detaches and falls off, which is thought to be a response to predators.[4] Because of this, typical rodent restraint methods cannot be used in these species, which can make the administration of parenteral medications difficult.

Due to their use in tissue regeneration research, these animals often undergo painful procedures, requiring the use of appropriate analgesic therapy. Some procedures performed in spiny mice include full-thickness skin wound creation, left anterior descending coronary artery ligation, hemi-crush injury to the spinal cord, and unilateral ureteral ligation.[5] Adequate analgesia is imperative in research animals to maintain animal welfare and comply with regulatory requirements. In addition, untreated pain can cause variability and skew experimental results.[6] Assessment of pain is notoriously difficult in laboratory rodents as prey species. In addition, many currently available assays likely oversimplify pain assessment, which may prevent determination of adequate analgesic protocols.[7,8] This is even more likely in spiny mice due to the limited research available on analgesic dosing in these species. There is a current need for effective pain relief medications in these species that account for their associated handling difficulties and susceptibility to skin sloughing.

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Long-acting analgesic therapies are becoming more popular among researchers using common laboratory rodents such as mice and rats. Long-acting medications allow for less frequent dosing and consequently less frequent handling, which may reduce stress and prevent injury in both the animals and their handlers.[6] Ethiqa XR and buprenorphine SR (Bup-SR) are 2 sustained-release buprenorphine products that have demonstrated analgesic efficacy in both mice and rats.[9–15] These products differ in terms of formulation (Ethiqa utilizes a lipid delivery system, while Bup-SR contains a polymeric vehicle) and FDA indexing (Ethiqa is FDA-indexed for subcutaneous use in mice and rats, whereas Bup-SR is compounded for off-label use).[6] A transdermal buprenorphine product developed and marketed for cats, Zorbium, has also been studied for use in mice and rats.[16–18] Zorbium has been shown to provide up to 96 hours of pain relief in mice,[17] and it was found to increase thermal withdrawal latency when given to rats.[16] Its route of administration avoids injections and associated restraint. Based on their demonstrated efficacy in other rodent species, these 3 products have the potential to be effective treatment options for spiny mice requiring pain relief.

This study aimed to evaluate the analgesic efficacy of 3 long-acting buprenorphine products—lipid-bound extended-release buprenorphine (XRB), polymeric sustained-release buprenorphine (SRB), and long-acting transdermal buprenorphine (TB)—in the Cairo spiny mouse (*Acomys cahirinus*). This was assessed by performing a thermal nociception assay (tail flick test) and noncompartmental pharmacokinetic analysis. We hypothesized that all 3 products would provide analgesia and plasma buprenorphine concentrations >1 ng/mL for up to 72 hours postadministration.

Materials and Methods

Ethical review. All procedures were performed at a facility accredited by AAALAC International and with the approval of the University of Michigan IACUC in accordance with the *Guide for the Care and Use of Laboratory Animals*.

Experimental animals. A total of 195 healthy male and female adult Cairo spiny mice (*A. cahirinus*) aged ≥2.5 months (range, 11-86 weeks) were included in this study. Their body weights ranged from 41 to 77 g. These were wild-type animals recycled from an in-house breeding program at the University of Michigan. Exclusion criteria included involvement in prior studies, previously documented pain, previous analgesic administration, and tail abnormalities.

The spiny mouse colony was designated specific-pathogen free for ectromelia virus, murine rotavirus A, Theiler murine encephalomyelitis virus, hantavirus, lymphocytic choriomeningitis virus, mouse adenovirus, mouse hepatitis virus, *Mycoplasma pulmonis*, mouse parvovirus, minute virus of mice, polyomavirus, pneumonia virus of mice, reovirus, Sendai virus, fur mites, and pinworms.

The spiny mice were housed in ventilated standard no. 3 rat cages (140 in^2; Allentown, Allentown, NJ) on Pure-o'Cel bedding (The Andersons, Maumee, OH) under constant environmental conditions (70-78 °F [21.1-25.6 °C], 30%-70% relative humidity, 12-hour light/12-hour dark cycle). Spiny mice were provided ad libitum rodent chow (2014 Teklad Global 14% protein rodent maintenance diet; Inotiv, West Lafayette, IN) in ceramic bowls and reverse osmosis water via a Lixit. They were housed in same-sex groups of 3 or 4, and they received nesting enrichment.

Randomization and blinding. Due to the limited availability of *Acomys* mice, animals were not randomized into treatment groups and each treatment was evaluated at separate times when animals became available through an internal recycling program. Confounders were not controlled in this study. Each treatment and control group underwent testing at separate times. One individual performed testing due to its proficiency with the nociception procedures, so blinding was not performed during group formation or the conduct of experiments. However, the tail flick analgesia meter used for this study is automated and the light beam turns off automatically when the test animal moves its tail, which may reduce potential bias.

Experimental procedures. ***Study design.*** Spiny mice were allocated into 3 treatment groups—XRB (Ethiqa XR; Fidelis Animal Health, North Brunswick, NJ) (n = 42; 21 females, 21 males), SRB (Bup-SR; Wedgewood Pharmacy, Swedesboro, NJ) (n = 42; 21 females, 21 males), and TB (Zorbium; Elanco, Indianapolis, IN) (n = 48; 24 females, 24 males). The spiny mice were briefly anesthetized for 1-2 minutes with 5% inhaled isoflurane (MWI Animal Health, Boise, ID), and the assigned treatment was administered as a single dose at baseline.

XRB was administered via subcutaneous intrascapular injection at a dose of 3.25 mg/kg using a 22-gauge needle per the manufacturer's recommendations. The dose used was that recommended for mice per the product label. SRB was administered via subcutaneous intrascapular injection at a dose of 1 mg/kg using a 22-gauge needle, and the injection site was massaged per the manufacturer's recommendations. This dose was chosen based on previous publications reporting a dose range of 0.6-1.5 mg/kg in mice.[6,19] TB was applied topically to the skin of the intrascapular region after parting the fur at a dose of 20 mg/kg. This dose was determined based on allometric scaling and previous publications in mice[17,18]; a midrange dose of 20 mg/kg was used. The contents of one 0.4-mL or 1-mL TB tube were emptied into a sterile screwcap microcentrifuge tube and withdrawn using a standard pipette. Spiny mice receiving TB were housed individually for 30 minutes after administration to allow the drug to dry prior to placing them back with their cagemates.

Spiny mice underwent tail flick testing at a set timepoint (1, 2, 4, 8, 24, 48, or 72 hours) posttreatment—an additional timepoint was added at 96 hours postadministration in the TB group due to promising tail flick results and the reported duration of action in laboratory mice.[17] Tail flick testing at the assigned timepoint was immediately followed by a blood collection performed via cardiac puncture under deep anesthesia with inhaled isoflurane. Animals were then humanely euthanized via an intracardiac injection of pentobarbital sodium and phenytoin sodium solution (Euthasol; Virbac, Carros, France).

Additional spiny mice were used for repeated serial tail flick testing. These spiny mice received XRB (n = 9; 4 females, 5 males), SRB (n = 9; 5 females, 4 males), or TB (n = 6; 3 females, 3 males) and underwent repeated tail flick tests at baseline and 1, 2, 4, 8, 24, 48, and 72 hours posttreatment. The treatments were administered as detailed previously. The associated saline control group contained spiny mice that either received a subcutaneous injection of saline in the intrascapular region using a 22-gauge needle (XRB/SRB controls, n = 10; 3 females, 7 males) or topical application of saline to the skin of the intrascapular region after parting the fur (TB controls, n = 6; 3 females, 3 males) and underwent tail flick tests at baseline and 1, 2, 4, 8, 24, 48, and 72 hours after saline administration. Animals were then anesthetized with inhaled isoflurane and humanely euthanized via an intracardiac injection of pentobarbital sodium and phenytoin sodium solution.

Plasma sample preparation and liquid chromatography–tandem MS methodology. Terminal blood collection was performed in treated animals to obtain samples for pharmacokinetic analysis to determine plasma buprenorphine concentrations at each nociception timepoint, as well as in untreated control animals (n = 18) to collect plasma for species-specific pharmacokinetic assay development. Approximately 0.5-1 mL of blood was collected prior to euthanasia, placed in lithium heparin microtainers, and centrifuged at 4 °C and 4,000 × g for 10 minutes. Plasma was then aliquoted into cryogenic vials and stored at −80 °C. Pharmacokinetic analysis was performed by the Pharmacokinetic and Mass Spectrometry Core at the University of Michigan College of Pharmacy (Ann Arbor, MI) using liquid chromatography–tandem MS.

A liquid chromatography–tandem MS method was developed for the quantification of buprenorphine using an AB SCIEX QTRAP 4500 mass spectrometer operated in positive electrospray ionization mode. Chromatographic separation was achieved on an XBridge C18 column (5 cm × 2.1 mm interior diameter, 3.5 μm) with a gradient elution using mobile phase A (0.1% formic acid in purified deionized water) and mobile phase B (0.1% formic acid in acetonitrile) at a flow rate of 0.4 mL/min. The injection volume was 10 μL, and the total run time was 5.6 minutes. The gradient program was as follows: 95% A at 0.5 minute, 5% A at 1.5 minutes, held until 3.5 minutes, returned to 95% A at 3.6 minutes, and reequilibrated until 5.6 minutes. Multiple reaction monitoring was used with the following transitions: m/z 468.20 → 414.3 for buprenorphine and m/z 472.20 → 414.3 for the internal standard (buprenorphine-d4). The decluttering potential, collision energy, collision exit potential, and electrode potential were optimized as follows: 89.0, 44.8, 5.6, and 9.6 V for buprenorphine and 99.0, 41.0, 30.0, and 8.0 V for the internal standard, respectively. Data acquisition and quantification were performed using Analyst software, with calibration curves constructed by plotting the peak area ratios of buprenorphine to the internal standard against known concentrations. The limit of quantification was 0.1 ng/mL.

The first group of untreated control animals, which were cohoused with treated animals, were found to have plasma buprenorphine concentrations >0 ng/mL, suggestive of contamination or allogrooming of topical buprenorphine products. Blood collection and pharmacokinetics were repeated in a new group of untreated control spiny mice that were not in contact with treated spiny mice.

Pharmacokinetic data analysis. A plasma buprenorphine concentration of 1 ng/mL was used as the suspected therapeutic concentration based on previous human and rodent studies.[20] No previous publications have evaluated plasma buprenorphine concentrations in spiny mice; however, plasma levels ranging from 0.5 to 1 ng/mL are considered antinociceptive in other laboratory rodents.[6]

A noncompartmental analysis was used to analyze pharmacokinetic data (PKanalix 2024R1, Simulations Plus, https://doi.org/10.5281/zenodo.11401684). Plasma concentrations below the limit of quantification were included as 0. The linear trapezoidal rule was used for the AUC calculation. The following pharmacokinetic parameters were estimated: C_{max}, T_{max}, $t_{1/2}$ obtained from $\ln(2)/\lambda_z$ (first-order rate constant estimated by linear regression of plasma concentration–time curve at terminal time points), AUC extrapolated to infinity ($AUC_{0\text{-}inf}$), % extrapolated AUC, and mean residual time. Statistical sampling of pharmacokinetic data was not performed.

Thermal nociception testing. Tail flick testing was performed to evaluate nociception following administration of an analgesic therapy or a control substance. All spiny mice were acclimated to the room and handled for 2 days prior to baseline testing. Prior to baseline or postanalgesic testing, the animals were acclimated to the room for 30-60 minutes and acrylic tubular rodent restrainers (IITC/Life Science, Woodland Hills, CA) for 5-10 minutes before tail flick testing. If spiny mice did not enter the restrainer on their own, they were briefly sedated with 5% inhaled isoflurane for 1-2 minutes, placed in the restrainer, and allowed to recover for ~5 minutes (until they were ambulatory inside the restrainer) prior to acclimation. The tail flick test setup used in this study is shown in Figure 1.

All tail flick testing was done using the IITC Tail Flick Analgesia Meter (IITC/Life Science, Woodland Hills, CA). Active intensity was established for each mouse by determining a mean basal latency of 2-4 seconds. All testing began at 45% active intensity 1 cm from the tail tip. Animals above the 4-second latency were retested in the same manner, increasing active intensity by 5% until a basal latency of 2-4 seconds was established. The established active intensity for each animal was maintained for all subsequent tests. The maximum cut-off time was 12 seconds to prevent tissue injury.

Each animal was tested 3 times, with an interval of 2-3 minutes between tests. Testing was performed 1, 1.5, and 2 cm from the tip of the tail. Latency was determined using the IITC Tail Flick Analgesia Meter's built-in timer to measure the time from light beam application to the animal flicking its tail to avoid the thermal stimulus. Percentage of maximum possible efficacy (%MPE), a measure of nociception, was calculated from the tail flick latency at each timepoint using the following formula: %MPE = [(latency at timepoint − latency at baseline)/(cut-off time − latency at baseline)] × 100.

The average %MPE value at each timepoint was then calculated to assess changes over time in each group and allow for comparison between groups. The mean latency and %CV ranges for all timepoints from 1 to 72 hours and for each group were also calculated.

Outcome measures. The outcome measures assessed in this study included changes in %MPE during acute thermal nociception tail flick latency and plasma buprenorphine concentrations at designated time intervals between treatment groups. Additional outcome measures included repeated measurement %MPE over time for the treatment and control groups.

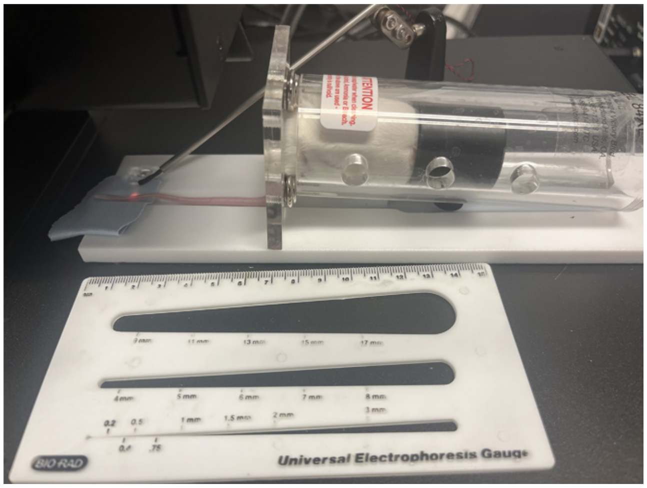

Figure 1. Tail Flick Test Setup Used in this Study. The test subject is placed in an acrylic rodent restrainer, and a ruler is used to measure the distance from the tail tip to the intended location of light beam application. Note that the image shows a CD-1 mouse.

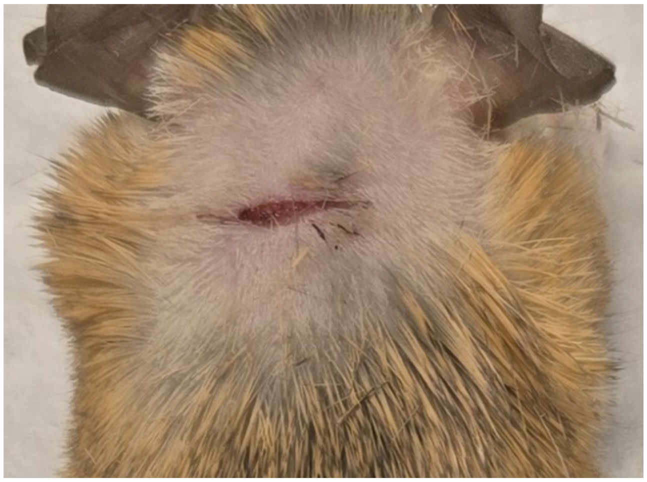

Figure 2. Skin Lesion at the Injection Site in a Female Spiny Mouse from the 24-h Polymeric Sustained-Release Buprenorphine Group. Surrounding hair has been shaved to aid visualization.

Statistical methods. GraphPad Prism version 10.4.1 (GraphPad Software, Boston, MA) was used for all statistical analyses. A mixed-effects model (treatment and time) was conducted with a Tukey post hoc correction for multiple comparisons to compare tail flick thermal latencies between treatment groups and timepoints after drug administration. All ANOVAs included the Geisser-Greenhouse correction to assist with any unequal variances between groups. A P value of <0.05 was considered significant.

Sample size was chosen based on previous publications in mice and with a larger a priori anticipated effect size,[21] as this study was designed for initial screening to test efficacy and functionality. A power analysis using $\alpha = 0.05$, power = 0.8, and an effect size of 1 indicated a sample size of 5 mice per group. Larger group sizes may be needed for more robust analysis to discriminate between different doses or effects.

Results

SRB-associated skin lesions. Several spiny mice who received SRB (21 out of 51 animals; 41%) developed mild to moderate ulcerative skin lesions at the injection site within 24 hours of administration (Figure 2). These lesions were predominantly noted at the 24, 48, and 72 hour timepoints. There was no apparent difference between sexes in terms of lesion development, as 10 out of 25 male spiny mice and 11 out of 26 female spiny mice were affected. No apparent signs of pain or distress were noted in these animals.

Pharmacokinetic analysis. Plasma buprenorphine concentrations were measured in female and male spiny mice receiving XRB, SRB, and TB (n = 3 females and 3 males per treatment group) at 1, 2, 4, 8, 24, 48, and 72 hours, as well as an additional timepoint at 96 hours for the TB group (Table 1). Two outliers were identified and not included in any analyses—a female spiny mouse in the 72 hours TB group with a plasma concentration of 83.7 ng/mL (the other 2 plasma concentrations were 1.02 and 0.42 ng/mL) and a female spiny mouse in the 8 hours SRB group with a plasma concentration of 324 ng/mL (the other 2 plasma concentrations were both 2.61 ng/mL). These outliers were outside the 99% CI at 126- and 195-fold the SD of the mean of the other 2 values at the same timepoint, respectively. High plasma concentrations were confirmed by reassay and review of liquid chromatography–tandem MS methodology and instrument quality control.

The XRB group displayed the longest duration at which mean plasma buprenorphine concentrations were >1 ng/mL—at least 72 hours in females and 48 hours in males. In the TB group, mean plasma buprenorphine levels remained >1 ng/mL for at least 24 hours in females and 48 hours in males. Lastly, both female and male spiny mice receiving SRB exhibited mean plasma buprenorphine levels >1 ng/mL for at least 8 hours postadministration.

A noncompartmental analysis was performed for female and male spiny mice receiving XRB, SRB, and TB (Table 2). The mean $AUC_{0\text{-inf}}$ was highest overall in the XRB group (female, 410 h·ng/mL; male, 282 h·ng/mL) followed by the TB (female, 305 h·ng/mL; male, 218 h·ng/mL) and SRB (female, 71 h·ng/mL; male, 41 h·ng/mL) groups. The mean C_{max} of buprenorphine was highest for spiny mice receiving XRB (female, 10.5 ng/mL; male, 8.5 ng/mL) followed by those receiving TB (female, 7.4 ng/mL; male, 7.5 ng/mL) and those in the SRB group (female, 3.4 ng/mL; male, 2.7 ng/mL). The drug $t_{1/2}$ was longest in spiny mice receiving TB (female, 42 hours; male, 29 hours), followed by XRB (female, 24 hours; male, 21 hours) and SRB (female, 12 hours; male, 13 hours).

Tail flick assay results in relation to plasma buprenorphine concentrations. The mean %MPE values were placed into dual-axis graphs with plasma buprenorphine concentrations for each treatment group to visualize the relationship between nociceptive and pharmacokinetic data (Figure 3).

In female spiny mice receiving XRB, mean plasma buprenorphine concentrations peaked at 8 hours, whereas the mean %MPE did not peak until 24 hours. Both values decreased to near baseline by 48 hours. Similar results were found in male spiny mice receiving XRB, in which mean plasma concentrations

Table 1. Plasma Buprenorphine Concentrations in Female and Male Spiny Mice Receiving XRB, SRB, and TB at 1, 2, 4, 8, 24, 48, and 72 h Postadministration

	XRB female		XRB male		SRB female		SRB male		TB female		TB male	
Time (h)	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1	2.6	1.1	3.9	2.4	2.2	0.9	2.7	0.9	2.2	0.8	5.8	5.3
2	6.7	2.7	5.6	3	2.7	0.7	1.7	0.4	6.5	6.4	4.3	1.8
4	9.2	1.1	8.5	0.9	3.4	2	1.7	0.8	7.4	1.7	6.1	1.2
8	10.5	3.4	6.1	1.4	2.6	0	1.7	0.8	4.4	0.6	7.5	0.2
24	6.2	1.9	5.6	1.9	0.8	0.7	0.4	0.2	3.9	1.4	2.2	0.1
48	1.9	0.6	1.3	0.4	0.4	0.08	0.1	0.2	0.8	0.6	1.1	0.7
72	1.8	0.9	0.9	0.3	0.06	0.1	0.05	0.09	0.7	0.4	0.8	0.1
96	–	–	–	–	–	–	–	–	1.6	1.9	0.4	0.1

An additional timepoint at 96 h was added for TB only. Data are presented as the mean plasma concentration (ng/mL) and SD. Abbreviations: SRB, polymeric sustained-release buprenorphine; TB, long-acting transdermal buprenorphine; XRB, lipid-bound extended-release buprenorphine.

Table 2. Noncompartmental Analysis of Female and Male Spiny Mice Receiving XRB, SRB, and TB

Group	$AUC_{0\text{-inf}}$ (h·ng/mL)	% Extrapolated AUC	C_{max} (ng/mL)	T_{max} (h)	$t_{1/2}$ (h)	$MRT_{0\text{-inf}}$ (h)
XRB female	400	16	10.5	8	24	36
XRB male	282	10	8.5	4	21	30
SRB female	71	1.5	3.4	4	12	18
SRB male	41	2.5	2.7	1	13	16
TB female	305	31	7.4	4	42	69
TB male	218	8	7.5	8	29	32

Values below the limit of quantification were replaced by 0. Data are reported as the mean value.
Abbreviations: $AUC_{0\text{-inf}}$, AUC extrapolated to infinity; $MRT_{0\text{-inf}}$, mean residual time extrapolated to infinity; SRB, polymeric sustained-release buprenorphine; TB, long-acting transdermal buprenorphine; XRB, lipid-bound extended-release buprenorphine.

peaked at 4 hours, the mean %MPE peaked at 24 hours, and both values had decreased by 48 hours (Figure 3A).

For both female and male spiny mice receiving SRB, the mean %MPE peaked at 8 hours postadministration and decreased below baseline by 48 hours in females and 24 hours in males. Mean plasma buprenorphine concentrations were highest at 4 hours in female spiny mice and 1 hour in male spiny mice receiving SRB (Figure 3B).

The mean %MPE in the TB treatment group peaked in female spiny mice at 24 hours postadministration and in male spiny mice at 8 hours. The mean %MPE remained above baseline for 48 hours in females and 72 hours in males. Mean plasma buprenorphine concentrations were highest in females at 4 hours and in males at 8 hours (Figure 3C).

Serial tail flick assay. For spiny mice undergoing serial tail flick assays at 1, 2, 4, 8, 24, 48, and 72 hours that received XRB, SRB, TB, or saline, 2-way ANOVA revealed no significant differences between females and males in the same treatment group and timepoint. Statistical comparisons between baseline tail flick latencies did not reveal any significant differences between treatment groups (females: XRB, 3.8 ± 1.2 seconds; SRB, 3.1 ± 0.7 seconds; TB, 3.3 ± 0.5 seconds; saline, 3.2 ± 0.5 seconds; males: XRB, 4.1 ± 1.2 seconds; SRB, 3.8 ± 1.2 seconds; TB, 3.5 ± 0.4 seconds; saline, 3.9 ± 1.2 seconds). One-way ANOVA demonstrated no significant difference between male and female baseline latencies. Comparisons of female and male spiny mice were not performed between treatment groups. The mean latency and %CV ranges for all timepoints from 1 to 72 hours for the saline group were as follows: females (latency, 3.14-3.86 seconds; %CV, 7.6%-27.6%) and males (latency 3.5-4.17 seconds; %CV 22%-29%). Two spiny mice intended for serial tail flick testing were removed from the study due to tail lesions and subsequent self-injury (1 male in the XRB group and 1 female in the TB group).

The mean %MPE was highest at all timepoints in spiny mice receiving TB, regardless of sex, and these values were elevated above baseline for the entire 72 hours in both female and male spiny mice (Figure 4). %MPE values were significantly higher than those of the saline control group at 1, 8, and 24 hours after administration in male spiny mice ($P = 0.003$, 0.008, and 0.0001, respectively) and 1, 2, 8, and 24 hours in female spiny mice ($P = 0.038$, 0.016, 0.001, and 0.048, respectively). Differences in %MPE between the TB and saline control groups were not significant at 2, 4, 48, and 72 hours in male spiny mice ($P = 0.214$, 0.08, 0.265, and 0.216, respectively) or at 4, 48, and 72 hours in female spiny mice ($P = 0.058$, 0.203, and 0.420, respectively). The mean latency and %CV ranges for all timepoints from 1 to 72 hours for the TB group were as follows: females (latency, 5.62-9.82 seconds; %CV, 7.8%-18.4%) and males (latency, 6.59-9.53 seconds; %CV, 3.7%-34.4%).

The mean %MPE was second highest in spiny mice receiving XRB, and these values were elevated above baseline for 72 hours postadministration in female spiny mice only. No significant differences in %MPE were found between the XRB group and the saline control group at any timepoint in either sex (females: $P = 0.447$, 0.241, 0.137, 0.114, 0.118, 0.071, and 0.223 at 1, 2, 4, 8, 24, 48, and 72 hours, respectively; males: $P = 0.053$, 0.174, 0.144, 0.093, 0.240, 0.236, and 0.361 at 1, 2, 4, 8, 24, 48, and 72 hours, respectively). The mean latency and %CV ranges for all timepoints from 1 to 72 hours for the XRB group were as follows: females (latency, 4.81-8.11 seconds; %CV, 19.4%-38.9%) and males (latency, 4.91-8.2 seconds; %CV, 26.8%-36.2%).

The mean %MPE was elevated above baseline for up to 48 hours in female and male spiny mice receiving SRB. %MPE values were significantly higher than those of the saline control group in male spiny mice at 1, 4, 8, 24, and 48 hours postadministration ($P = 0.017$, 0.010, 0.014, 0.039, and 0.008, respectively); however, no significance was achieved in male spiny mice at 2 or 72 hours ($P = 0.174$ and 0.998, respectively) or in the female spiny mice receiving SRB at any timepoint ($P = 0.528$, 0.561, 0.286, 0.295, 0.709, 0.999, 0.178 at 1, 2, 4, 8, 24, 48, and 72 hours, respectively). The mean latency and %CV ranges for all timepoints from 1 to 72 hours for the SRB group were as follows: females (latency, 3.34-5.09 seconds; %CV, 13%-47.5%) and males (latency, 3.25-6.67 seconds; %CV, 7.4%-29.2%).

%MPE was also compared between treatments. Significant differences in %MPE between the TB and SRB groups were noted at 8 hours ($P = 0.041$) in male spiny mice and at 2 ($P = 0.017$) and 8 hours ($P = 0.007$) in female spiny mice (Figure 4). No significant differences in %MPE were observed between the XRB and SRB groups in female or male spiny mice.

Discussion

This study aimed to assess the analgesic efficacy and pharmacokinetics of 3 long-acting buprenorphine formulations in spiny mice. There is currently little to no information available on analgesic dosing in spiny mice. In a clinical setting, analgesic therapy selection and administration in *Acomys* mice is therefore based on data from other rodent species such as mice and rats. Both XRB and TB displayed sufficient duration and efficacy to be used as analgesic therapies in Cairo spiny mice. Although XRB demonstrated a longer duration of action on pharmacokinetic analysis, TB exhibited greater increases in %MPE. When selecting an analgesic regimen for spiny mice, we recommend considering the individual animal to determine whether ease of application or duration of action is the main priority, as TB can be easily applied to conscious animals but XRB may last longer in the circulation based on slightly higher plasma concentrations at 72 hours.

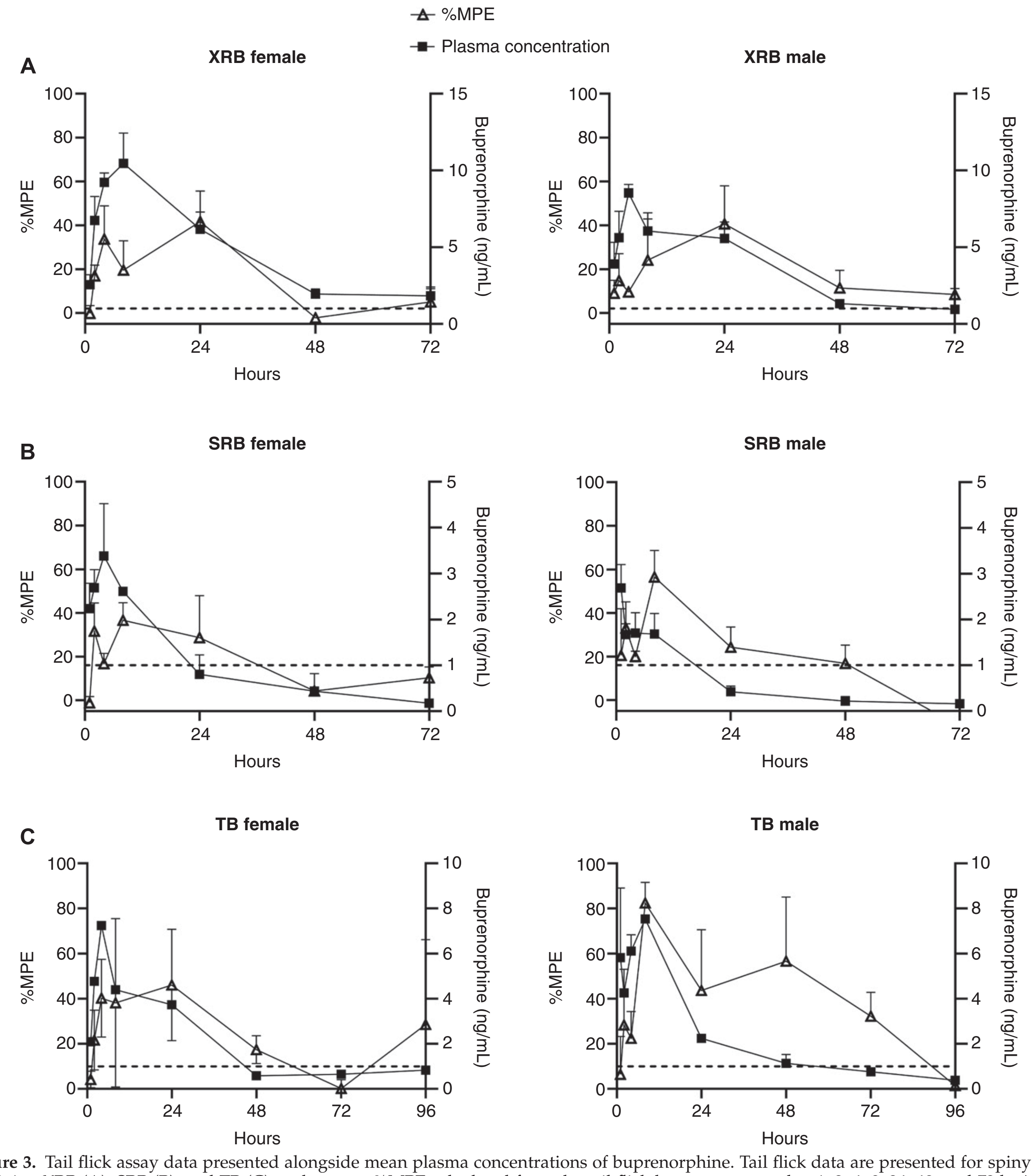

Figure 3. Tail flick assay data presented alongside mean plasma concentrations of buprenorphine. Tail flick data are presented for spiny mice receiving XRB (A), SRB (B), and TB (C) as the mean %MPE calculated from the tail flick latency measured at 1, 2, 4, 8, 24, 48, and 72 h. An additional timepoint at 96 h was added for TB only. n = 3 female and male spiny mice per timepoint (except groups with excluded outliers). The dotted line represents a plasma buprenorphine concentration of 1 ng/mL. %MPE, percentage of maximum possible efficacy; SRB, polymeric sustained-release buprenorphine; TB, long-acting transdermal buprenorphine; XRB, lipid-bound extended-release buprenorphine.

On pharmacokinetic analysis, XRB demonstrated the longest duration at which mean plasma buprenorphine concentrations were >1 ng/mL—72 hours in female spiny mice and 48 hours in male spiny mice. In the TB group, mean plasma buprenorphine levels remained >1 ng/mL for 48 hours in males and 24 hours in females, although the concentration increased above this level again at 96 hours in female animals only. This discrepancy may be due to differences in absorption, elimination, or clearance mechanisms between male and female mice[22] administered transdermal buprenorphine. SRB maintained mean plasma buprenorphine concentrations >1 ng/mL for 8 hours in both sexes. Importantly, note, however, that when the SD was considered in addition to the mean plasma concentration, levels fell below the threshold of 1 ng/mL at an earlier timepoint in most treatment groups. This suggests that if 1 ng/mL is considered the therapeutic threshold, some animals fell below this level and may not have received adequate analgesia despite a seemingly sufficient mean value, which should be considered by spiny mouse users performing painful procedures and veterinary staff attempting to provide pain relief.

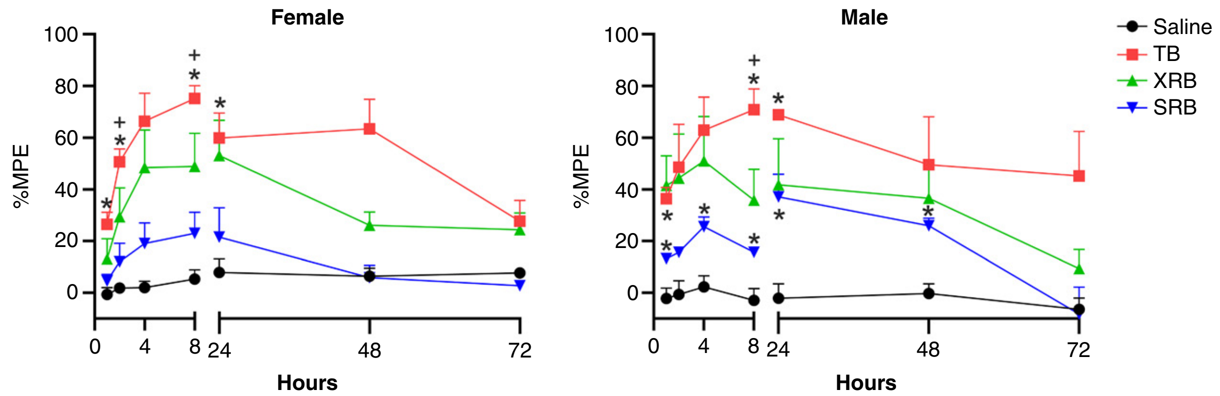

Figure 4. Serial tail flick data. Data from female and male spiny mice receiving XRB, SRB, TB, and saline (control) are presented as the %MPE calculated from the tail flick latency measured at 1, 2, 4, 8, 24, 48, and 72 h. Data are presented as the mean and SD. $^{*}P < 0.05$ between treatment and saline control; $^{+}P < 0.05$ between TB and SRB. %MPE, percentage of maximum possible efficacy; SRB, polymeric sustained-release buprenorphine; TB, long-acting transdermal buprenorphine; XRB, lipid-bound extended-release buprenorphine.

On the noncompartmental analysis, the XRB group had the highest overall $AUC_{0\text{-inf}}$ and C_{max}, followed by TB, indicating greater systemic exposure to the drug and more complete absorption, respectively. Although the AUCs of XRB and TB were relatively similar between males and females, the XRB group maintained longer plasma concentrations. The SRB group had the lowest $AUC_{0\text{-inf}}$ and C_{max} values, which could potentially suggest a reduced rate of absorption or rapid clearance of this drug in *Acomys* mice. Since SRB was associated with injection site lesions, the rate of absorption may have affected AUC values to a larger extent. The TB group demonstrated the longest half-life compared with XRB and SRB. This may be due to delayed absorption via the topical route coupled with rapid elimination. An intravenous group was not included to detect whether flip-flop kinetics were observed in which the rate of elimination is greater than the rate of absorption. The higher plasma concentration (83.7 ng/mL) observed in the outlier from the female 72-hour TB group was most likely due to variances in transdermal drug absorption via transcellular diffusion through keratinocyte lipid bilayers, intracellular ceramide-cholesterol matrices, tight junctions, or follicular glands.[23,24] Such variations have been previously observed in other species[25] and in our prior experience (data not published). To accommodate for such variances, larger sample sizes and longer sampling intervals are recommended when performing future TB pharmacokinetic studies.

When assessing serial tail flick assays over time, the mean %MPE was higher in the XRB and TB groups than in the control group at all timepoints for 72 hours postadministration and in the SRB group for the first 24-48 hours. Overall, the mean %MPE was greatest at all timepoints in the TB group, followed by the XRB group, suggestive of total drug exposure, as these groups also had the highest AUC values or higher opioid receptor occupancy. Increased %MPE is indicative of analgesia; however, additional studies are warranted to determine whether the observed drug effect on nociception translates to clinical or therapeutic analgesia depending on the surgical procedure. Additional studies evaluating behavioral and spontaneous or evoked pain are warranted to best determine an accurate dose and duration. In addition, a direct correlation between buprenorphine %MPE and plasma concentration is complicated due to slow extravascular absorption into the CNS, sustained high-affinity receptor binding, and slow-release receptor equilibrium kinetics.[26] Previous literature has demonstrated a counterclockwise hysteresis with buprenorphine when transdermally administered.[27] The hysteresis is such that initial plasma concentrations produce a smaller effect on %MPE while maximum effect occurs at later timepoints, as seen in the male TB group. As a result, clinical observations for pain should be used in conjunction with plasma concentrations.

Additional analgesic efficacy studies and pharmacokinetic evaluations are needed to fully elucidate which plasma concentration is indicative of therapeutic analgesia in spiny mice. It is uncertain whether a therapeutic plasma buprenorphine concentration of 1 ng/mL directly correlates or can be applied to *Acomys* spp., yet the tail flick latencies and correlated plasma concentrations observed in this study tend to support analgesia near this level. Additional studies evaluating alternative doses of these buprenorphine formulations are also warranted to determine whether duration of action can be extended and to what extent duration is affected by dosage.

Several spiny mice receiving SRB developed mild to moderate ulcerative skin lesions at the injection site during the conduction of this study. In addition to the spiny mouse's predisposition to skin sloughing, possible factors that may have contributed to these lesions include the drug formulation, massaging of the injection site per manufacturer recommendations, and needle size, although the same needle size was used when administering XRB and saline, and lesions were not noted in these groups. SRB has been shown to cause skin lesions in other rodent species, including mice[28,29] and rats,[30] as well as other laboratory species such as dogs,[31] pigs,[32] and common marmosets.[33] XRB has been reported to cause subcutaneous nodules[9] and granulomatous reactions at the injection site in rats[13]; however, no apparent gross lesions were noted in the spiny mice receiving XRB in this study. It is unclear how translatable these findings in other rodents are to spiny mice, as *Acomys* mice have several differences from laboratory mice (*Mus musculus*) in terms of skin composition (eg, collagen content, myofibroblasts) and their ability to completely regenerate skin, hair follicles, and adipocytes.[3] Regardless, due to the poorer pharmacokinetic and nociceptive data in the SRB group as well as the skin lesions associated with SRB administration, we do not recommend the use of this drug in spiny mice without additional evaluation. This may involve investigating different doses of SRB or conducting studies in which the skin is not manipulated after SRB administration.

Several studies have evaluated the use of XRB in more common laboratory rodent species. Pharmacokinetic studies performed in various strains of mice have found this product to maintain plasma buprenorphine levels >1 ng/mL for 48-72 hours postadministration.[10–12,15] These findings are consistent with our results in that spiny mice given the recommended mouse dose of XRB (3.25 mg/kg) had mean plasma buprenorphine levels >1 ng/mL for 48 (males) to 72 hours (females). Perhaps the most relevant is a study performed in Mongolian gerbils (*Meriones unguiculatus*), the closest related common laboratory species to spiny mice,[34] which found that XRB maintained therapeutic plasma buprenorphine levels >1 ng/mL for at least 48 hours after administration when given at a 1 mg/kg dose.[35] Since the present study is the first (to our knowledge) to assess plasma buprenorphine concentrations and nociceptive assays in spiny mice, a therapeutic threshold cannot be conclusively defined in these species.

Two pharmacokinetic studies in laboratory mice have been published regarding the use of TB, which was approved by the FDA in January 2022 for the management of postoperative pain in cats.[36] The first study administered TB at a dose of 30-40 mg/kg to C57BL/6 mice and found that plasma buprenorphine levels remained >1 ng/mL for 72-96 hours.[17] Similarly, in cats, TB has been shown to form a depot in the skin and remain in circulation for up to 4 days.[25] The second murine study performed allometric dosing in CD-1 mice. A single dose of TB (9-13 mg/kg) maintained plasma buprenorphine levels >1 ng/mL for 24 hours postadministration, although repeated dosing at 24 and 48 hours was required to maintain plasma concentrations above this threshold for 72 hours.[18] It is possible that repeated administration of TB may help to maintain therapeutic levels of buprenorphine for extended periods of time in spiny mice based on our finding that mean plasma buprenorphine concentrations remained >1 ng/mL for 24 hours following application in female spiny mice receiving TB. The dose used in the present study (20 mg/kg) allowed for easy calculation of drug volume due to the concentration of TB (20 mg/mL), although this dose may be optimized in future studies and subsequently affect the duration of action. Simmons et al[18] also found that allogrooming occurred between dosed and nondosed cagemates. In the present study, the control pharmacokinetic results had to be repeated due to the presence of buprenorphine in a nondosed cagemate, suggesting that spiny mice also allogroom, which should be considered when selecting an analgesic therapy for group-housed spiny mice.

In the present study, 2 spiny mice were removed due to suspected self-induced tail injuries following the administration of XRB and TB, respectively. Spiny mice exhibit false caudal autotomy in which they shed their tail skin in response to predators.[3] They then gnaw any underlying tail tissue until it is removed. A field study performed over 25 months in Israel examined the incidence of partial or total tail loss in *A. cahirinus* and found that 12% of male spiny mice and 25% of female spiny mice had some degree of tail loss.[4] It could not be determined in our animals if their tail injuries were due to self-injury following drug administration or potential irritation experienced during repeated serial tail flick assays. It is possible that self-injurious behaviors were related to the administration of XRB and TB, as both drugs have been associated with these behaviors in rats[16]; alternatively, these injuries may have been unrelated to our study.

This study had several limitations. First, randomization into treatment groups and blinding were not performed due to limited availability of *Acomys* mice at our institution resulting in each treatment group undergoing experimentation at different times as spiny mice became available via a recycling program. In addition to this, only one species of spiny mice, *A. cahirinus*, was included in the study, and it is possible that our results may not translate to other species of spiny mice due to genetic variance. Potential discrepancies between female and male spiny mice were not thoroughly investigated due to the small sample size per timepoint, so no conclusions could be drawn in terms of sex differences. Lastly, when performing tail flick testing, some animals required brief sedation with isoflurane to place them in the restrainer. They were allowed to recover and acclimatize; however, it is possible that this could have affected our tail flick results compared with spiny mice that were more tolerant to handling and did not require any sedation.

Because this is the first study to our knowledge to assess analgesia in spiny mice, there are several potential future directions. First, only one dose of each drug was evaluated, and it is possible that their effects could be dose-dependent—future studies optimizing the dose of these medications are therefore warranted. In addition, this study evaluated nociception using a tail flick assay on healthy tail tissue, so future research can consider using a surgical model such as a plantar incision and alternative nociceptive assays such as the Von Frey and Hargreaves tests to assess mechanical and thermal hypersensitivity, respectively, to further assess nociception and analgesic therapies in these species. Only buprenorphine was examined in this study, but other opioids and additional classes of analgesics (eg, NSAIDs, local anesthetics) could be assessed in future studies. The present study mainly focused on nociceptive and pharmacokinetic data, so future research could also evaluate additional parameters following the administration of analgesics, including body weight assessments and behavioral assays. Because of the skin lesions noted with SRB and similar findings in previous studies evaluating SRB and XRB in other rodents, additional evaluation of the injection site via skin scoring and histology would also be beneficial to determine the effect of these drugs on the skin. Lastly, there was some evidence of allogrooming in our spiny mice given TB, as was seen in the murine study by Simmons et al,[18] so this could be examined further to determine the frequency in spiny mice and whether any related adverse effects are expected following topical administration.

In conclusion, XRB and TB were found to be promising analgesic therapies in Cairo spiny mice (*A. cahirinus*) based on pharmacokinetic analysis and nociceptive assays. Both products resulted in increased %MPE compared with baseline for 48-72 hours, suggesting a sustained duration of action. Animals receiving XRB maintained mean plasma buprenorphine levels >1 ng/mL for a longer duration of 48-72 hours, compared with 24-48 hours in those receiving TB. These findings may be translatable to other species of *Acomys* mice; however, this cannot be confirmed without additional studies. Treatment selection between XRB and TB should be determined on an individual basis, as XRB may remain in the circulation longer at potentially therapeutic levels while TB is easier to administer to conscious animals.

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Conflict of Interest

The authors have no conflicts of interest to declare.

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Author Contributions

All authors assisted in study design and data collection. L.M.D., P.A.L., C.D.F., and D.E.H. assisted with pharmacokinetic and data analysis. All authors assisted with manuscript preparation, review, and revisions.

Generalizability/translation

The findings of this study may be applied to *Acomys* spp. to augment a refinement to analgesia.

Protocol registration

This study was approved by the University of Michigan's IACUC.

Data availability

Data can be supplied with permission.

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